

Tomás Cabrera

(303) 519-3955 • tcabrera@andrew.cmu.edu
[tomas-cabrera.github.io](https://github.com/tomas-cabrera) • <https://www.linkedin.com/in/tomas-cabrera>

Education

CARNEGIE MELLON UNIVERSITY

Doctor of Philosophy in Physics
GPA: 4.0/4.0

Pittsburgh, PA, USA
Graduation Fall 2026

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

Bachelor of Science in Physics - Major; Astronomy Minor
GPA: 4.6/5.0

Cambridge, MA, USA
Graduation May 2019

Skills

Software Skills: Python (numpy, pandas, scipy, matplotlib, astropy), Unix, Slurm, git, GitHub; familiarity with C, SQL

Analytical Skills: Data visualization, Bayesian inference, statistics, N-body simulations

Astronomy Skills: Photometry, difference imaging, spectroscopy, observation planning/execution, transient classification, gravitational wave follow-up; HEALPix skymaps, HATS/LSDB, TOPCAT, DS9

Observation experience: DECam (15+ nights), Gemini GMOS (10 hours), SALT RSS (3+ targets), *Swift* (5+ targets), Lowell Discovery Telescope (4 hours)

PhD Research

CARNEGIE MELLON UNIVERSITY

Constraining dynamical formation channels of LIGO-Virgo-KAGRA binary black holes

Pittsburgh, PA, USA
2020-present

Real-time discovery of electromagnetic counterparts to gravitational wave events

- Designed and optimized GPU-enabled Python image differencing pipeline to process DECam telescope images
- Developed software tools to assist scientists in real-time observation planning for short-lived phenomena, including the publication of a selection of tools as the [healpix-painter](#) Python package
- Installed and operated pipeline on Pittsburgh Supercomputing Center and NERSC systems to reduce nightly telescope observations within 12 hours of acquisition for rapid discovery of fading transients
- Initialized and populated SQL database to catalog pipeline products, and facilitate spatial crossmatching with existing datasets of 10^9+ objects to evaluate discoveries in greater astrophysical context
- Curated multi-wavelength and spectroscopic datasets of promising candidates for further classification

Measuring correlations between binary black holes and formation environments

- Refined and applied Bayesian inference methods to active galactic nuclei (AGN) catalogs and poorly-localized black hole mergers to constrain the fraction of AGN-hosted mergers
- Optimized legacy Monte Carlo N-body simulation codes in C language to study populations of black holes in star clusters, enabling simulations of 10^7 -body systems

AGN transient modeling with archival photometric data

- Reduced archival DECam deep-field data to generate 13-year baseline AGN lightcurves
- Collaborated with multidisciplinary personnel to develop methods of identifying and classifying AGN transients

Publications

MAJOR CONTRIBUTIONS

Cabrera, T., Palmese, A., Fishbach, M. "Multimessenger constraints on LIGO/Virgo/KAGRA gravitational wave binary black holes merging in AGN disks." *Astrophysical Journal*, 1000, 2. 2026. Print.

Hu, L., **Cabrera, T.**, et al. "A GPU-Accelerated Transient Detection Pipeline for DECam Time-Domain Surveys." Accepted for publication in *Publications of the Astronomical Society of the Pacific*. 2026. Print.

González Prieto, E., Rodríguez, C.L., **Cabrera, T.** "Growing the Intermediate-mass Black Hole in Omega Centauri." *The Astrophysical Journal Letters*, 990, 2. 2025. Print.

Hu, L., **Cabrera, T.**, et al. "Kilonova Constraints for the LIGO/Virgo/KAGRA Neutron Star Merger Candidate S250206dm: GW-MMADS Observations." The Astrophysical Journal Letters, 990, 2. 2025. Print.

Cabrera, T., Palmese, A., et al. "Searching for electromagnetic emission in an AGN from the gravitational wave binary black hole merger candidate S230922g." Astrophysical Journal, 953, 19. 2024. Print.

Cabrera, T., Rodriguez, C.L. "Runaway and hypervelocity stars from compact object encounters in globular clusters." Physical Review D, 110, 12. 2023. Print.

SIGNIFICANT CONTRIBUTIONS

O'Connor, B.,... **Cabrera, T.**, et al. "EP260321a/SN 2026gzf: The Faintest Shock Breakout Associated with a Broad-Lined Supernova." Submitted.

O'Connor, B.,... **Cabrera, T.**, et al. "GRB 260310A/SN 2026fgk: Photometric and Spectroscopic Evolution of a Nearby GRB-Supernova and an Exceptionally Bright Afterglow at $z=0.153$." Submitted.

Hall, X.,... **Cabrera, T.**, et al. "Electromagnetic Follow-up of the Sub-Solar Mass Gravitational Wave Candidate S251112cm: Kilonova Constraints and a Coincident IIb Supernova." Submitted.

O'Connor, B.,... **Cabrera, T.**, et al. "SN 2025adpq: A Type Ia supernova in a collisional ring formed during a major galaxy merger." Submitted.

Ahumada, T., ... **Cabrera, T.**, et al. "LIGO/Virgo/KAGRA neutron star merger candidate S250206dm: Zwicky Transient Facility observations." Publications of the Astronomical Society of the Pacific, 138, 3. 2026. Print.

O'Connor, B., ... **Cabrera, T.**, et al. "AT2025ulz and S250818k: Deep X-ray and radio limits on off-axis afterglow emission and prospects for future discovery." Astrophysical Journal Letters, 995, 2. 2025. Print.

Kasliwal, M., ... **Cabrera, T.**, et al. "ZTF25abjmnps (AT2025ulz) and S250818k: A Candidate Superkilonova from a Sub-threshold Sub-Solar Gravitational Wave Trigger." Astrophysical Journal Letters, 995, 2. 2025. Print.

Kunnumkai, K., ... **Cabrera, T.**, et al. "Detecting Electromagnetic Counterparts to LIGO/Virgo/KAGRA Gravitational-wave Events with DECam: Neutron Star Mergers" Astrophysical Journal, 993, 1. 2025. Print. Pillas, M., ... **Cabrera, T.**, et al. "Limits on the ejecta mass during the search for kilonovae associated with neutron star-black hole mergers: A case study of S230518h, GW230529, S230627c and the low-significance candidate S240422ed." Physical Review D, 112, 8. 2025. Print.

O'Connor, B., ... **Cabrera, T.**, et al. "Characterization of a Peculiar Einstein Probe Transient EP240408a: An Exotic Gamma-Ray Burst or an Abnormal Jetted Tidal Disruption Event?" The Astrophysical Journal Letters, 979, 2. 2025. Print.

16 NASA GCN Circulars and **779 TNS submissions** reporting transients found with the GW-MMADS pipeline

Conferences & Colloquia

"Of Needles and Haystacks: Lessons from the GW-MMADS follow-up campaign during the fourth LIGO/Virgo/KAGRA observing run." Throughput Computing Week, Madison, WI, USA. June 2026. **Invited talk.**

"When the darkness shines: The search for electromagnetic counterparts to gravitational wave events and its use as a probe of binary black hole formation." APS April Meeting, Sacramento, CA, USA. April 2024. Contributed talk.

"Rapid discovery of electromagnetic counterparts of gravitational waves with DECam." Unveiling the dynamical universe: Cosmic streams in the era of Rubin, Puerto Varas, Chile. December 2023. Contributed talk.

"Where big data become fast data: Challenges in the collection and use of multi-messenger data" NOIRLab South Colloquium, La Serena, Chile. September 2023. **Invited colloquium.**

"Runaway and Hypervelocity Stars from Compact Object Encounters in Globular Clusters" MODEST 23: Star Clusters in the Post-Pandemic Era. Evanston, IL, USA. August 2023. Contributed talk.

Leadership, Teaching, & Outreach

CARNEGIE MELLON UNIVERSITY

Graduate Mentor to Undergraduate Research

• Guided undergraduate students through projects as introduction to professional research

Pittsburgh, PA, USA
Fall 2024-present

<i>Physics Teacher's Program Assistant</i>	Summer 2025
• Prepared Jupyter notebooks introducing coding and modern physics methods at high-school levels, and iterated with local high school teachers to improve content	
<i>Astronomy on Tap Presenter</i>	April 2023
• Delivered public lecture on contemporary astronomy research	
<i>Teaching Assistant</i>	Fall 2020-Fall 2021, Spring 2026
• Led weekly recitations and answered questions on course material	
• Graded and provided feedback on homework assignments and midterms	

LSST DARK ENERGY SCIENCE COLLABORATION SPRINT WEEK	Pittsburgh, PA, USA
<i>Local Organizing Committee</i>	October 2023
• Organized and led after-hours activities, including evening social for junior members and museum visit	
• Served logistical needs including booking meeting spaces, distributing food, and setting up/tearing down rooms	

Allocations

Telescope

PI: "Riddles in the Dark: Deciphering the origin of BBH mergers through spectroscopic study of potential AGN flare counterparts" (Gemini Fast Turnaround January 2025, renewed May 2025, 6 hours)
PI: "Searching for variability in the spectrum of an electromagnetic counterpart candidate to the gravitational wave event S230922g" (Gemini DDT 2023B, 1 hour)
PI: "Spectroscopic follow-up of a new electromagnetic counterpart candidate to the gravitational wave event S230922g" (Gemini DDT 2023B, 1 hour)
co-I: "GW-MMADS: Gravitational Wave Multi-Messenger Astronomy DECam Survey" (PIs Andreoni & Palmese; DECam GW follow-up survey during the LIGO/Virgo/KAGRA O4 run, 2023-2025; proposal ID 2023B-851374, extended from allocated standard programs 2022B-715089, PI Palmese and 2022B-922046, PIs Andreoni & Palmese)
co-I on various additional electromagnetic counterpart discovery proposals for the LIGO/Virgo/KAGRA O4 run

Compute

PI: "Archival search for AGN transients in DECam deep fields" (Pittsburgh Supercomputing Center Discover Allocation, 2025-2026)
co-I on various PSC and NERSC allocations supporting PhD research since 2020

Workshops

ZWICKY TRANSIENT FACILITY SUMMER SCHOOL	Minneapolis, MN, USA
• Engaged in exercises in transient astronomy and related technologies	Summer 2023
INTERNATIONAL HIGH PERFORMANCE COMPUTING SUMMER SCHOOL	Athens, Greece
• Attended sessions on OpenMP, MPI, CUDA, and OpenACC parallelization models in C	Summer 2022
• Practiced instructed skills on provided HPC exercises	
• Received exposure to international HPC ecosystems and resources	

Honors & Awards

NSF GRFP HONORABLE MENTION (2022)

Extracurriculars

TOMÁS LIUS DE VICTORIA CHORAL CONDUCTING FELLOWSHIP

Pittsburgh Oratory Conducting Fellow

Pittsburgh, PA, USA
August 2023-May 2026

- Led rehearsals and conducted performances for 12-16 person choir, including works spanning Gregorian chant to Renaissance and modern polyphony

CHRIST IN THE CITY

Outreach Missionary

Denver, CO, USA
August 2019-July 2020

- Organized and led weekly park lunches by preparing and serving meals, assisting in the distribution of donated material goods, and managing volunteers
- Helped mentor and guide 300+ college students over 5 week-long mission trips

SOUTH BRONX EDUCATION FOUNDATION

Group Leader

New York, NY, USA
Summer 2019

- Tutored and mentored eight 2nd-grade boys in classroom setting to build good academic and personal habits
- Led weekly field trips to local attractions